Project Proposal

Movie Recommender System Using Item-based Collaborative Filtering

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1. Source of Your Project

Book: Python Machine Learning Workbook for Beginners by M. Usman Malik

Chapter/Project: Project 7 – Movie Recommender System Using Item – Based Collaborative Filtering (Pages 177-201)

2. Project Description

This project that I have chosen essentially develops a movie recommendation system using item-based collaborative filtering. The system analyzes ratings of movies from users and choses different relationships between movies based on corresponding rating patterns. Movies that tend to receive similar ratings from similar users will be identified, in which the model can then recommend new movies based on movies they have already seen and liked. This is already a pretty basic system used in many streaming platforms such as Netflix, Hulu, etc. as a way to personalize and improve the experience.

3. Libraries and Dataset

The libraries used for this project are Numpy, Pandas, Matplotlib, and Seaborn. The dataset used is The MovieLens Small Dataset which contains about 100k movie ratings, over 9k movies, and roughly 600 users. The files used are “movies.csv” and “ratings.csv”. The dataset altogether provides ratings that can allow similarities between movies to be calculated.

4. Data Preprocessing

There is multiple preprocessing steps that are preformed before creating the movie recommendation system. These steps include loading movie information & ratings datasets, removing unnecessary columns, merging movie & rating datasets, creating a user-movie rating matrix with rows for users and columns for movies, calculating correlations between movies based on rating patterns, and finally, filtering the movies with insufficient ratings to improve recommendation accuracy.

5. Model Used in the Project

This project uses item-based collaborative filtering which calculates correlations between movies, instead of predicting ratings with a machine learning classifier. For a movie to be considered related, they have to essentially receive similar ratings. The model uses Pearson Correlation, Collaborative Filtering Techniques, and User-Movie rating matrix. The engine uses the model to then suggest movies to the user based on strong correlations to ones that they've already liked.

6. End Goal

The desired outcome for this project is a fully functioning movie recommendation system that suggests movies to users, based off movies that they've previously liked and shared high ratings for. The final system should be able to accept multiple movies that the user enjoys, find similar movies by using collaborative filtering, and produce a ranked list of recommended movies. The end goal is to show how a recommendation system could be built using real-world rating data, that is then applied to personalized movie recommendations.